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eMotion-GAN: A Motion-based GAN for Photorealistic and Facial Expression Preserving Frontal View Synthesis

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Background & Motivation

Problem

Pose variation (especially non-frontal views) significantly degrades the performance of facial expression recognition and related tasks in real-world settings (e.g., healthcare, surveillance). Standard frontal view synthesis often introduces visual artifacts or distortion in expressions.

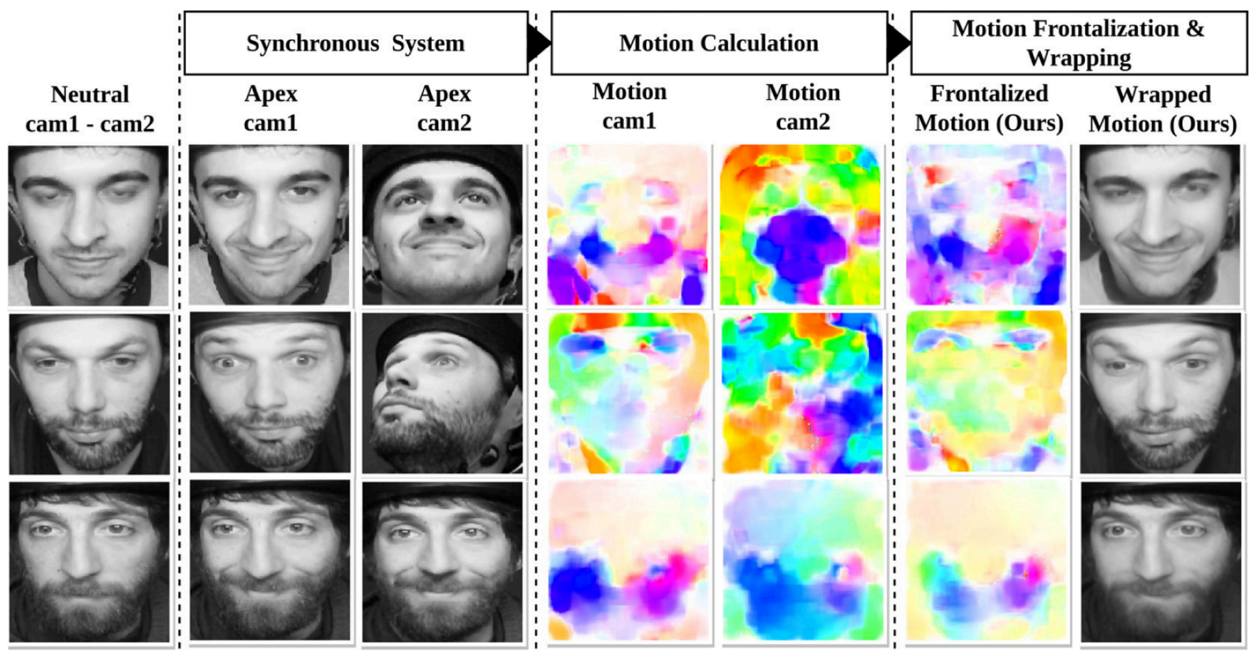
Goal

Develop a generative model that synthesizes high-quality frontal face images while preserving authentic facial expressions, even under large pose variations.

Key Insight

Instead of producing frontal images solely in the appearance (pixel) space, treat head pose as structured optical flow noise and separate motion into:

- Pose-related motion (undesired)
- Expression-related motion (desired)



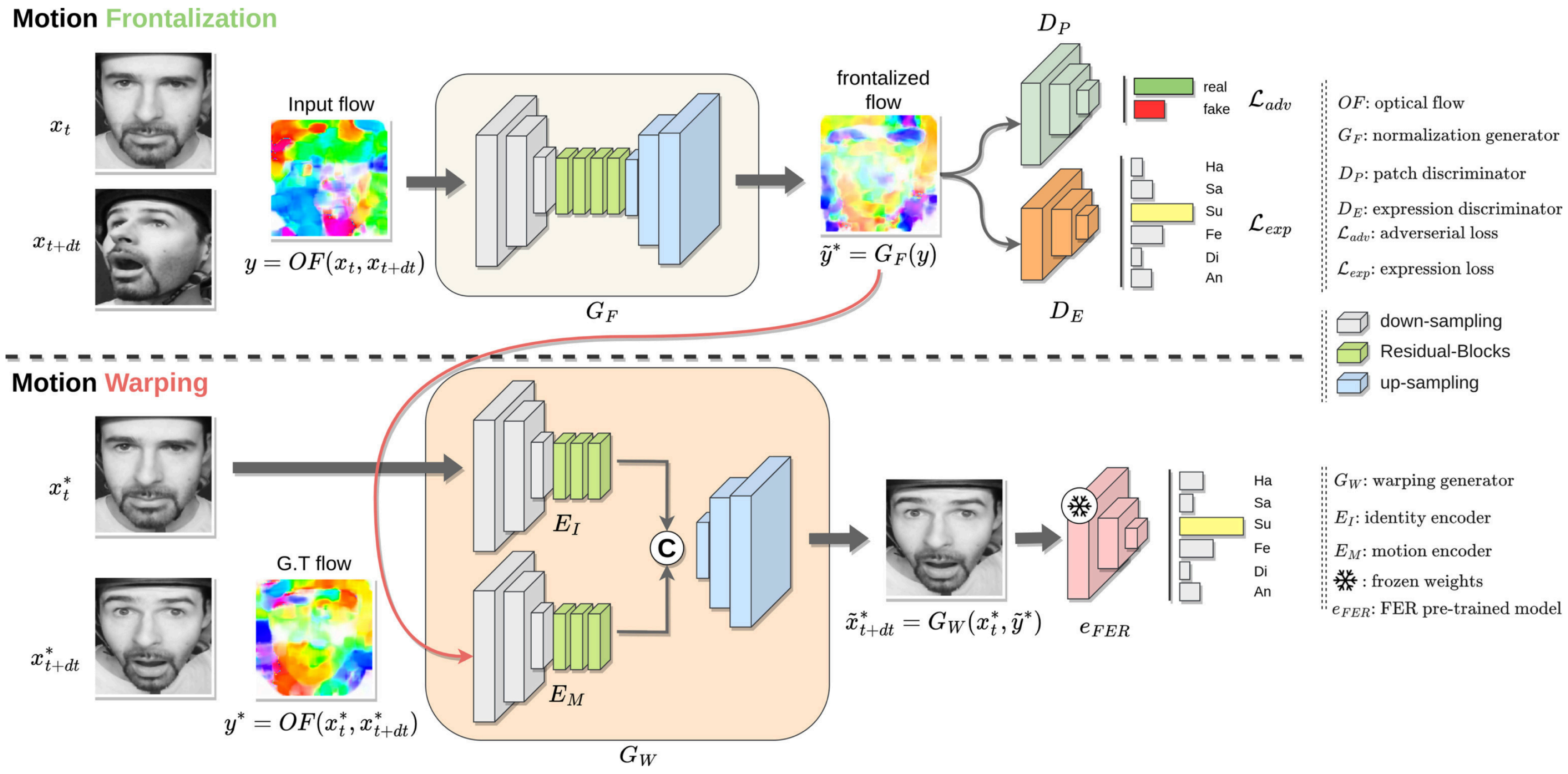
Proposed Method: eMotion-GAN

Architecture Overview

- Two-stage generative framework
  - Motion decomposition into pose and expression
  - Synthesis controlled to retain expression and remove pose effects
- No facial landmarks required
  - Robust to inaccurate or missing landmarks in input images
- Optical flow modeling
  - Treat structured motion explicitly rather than purely appearance features

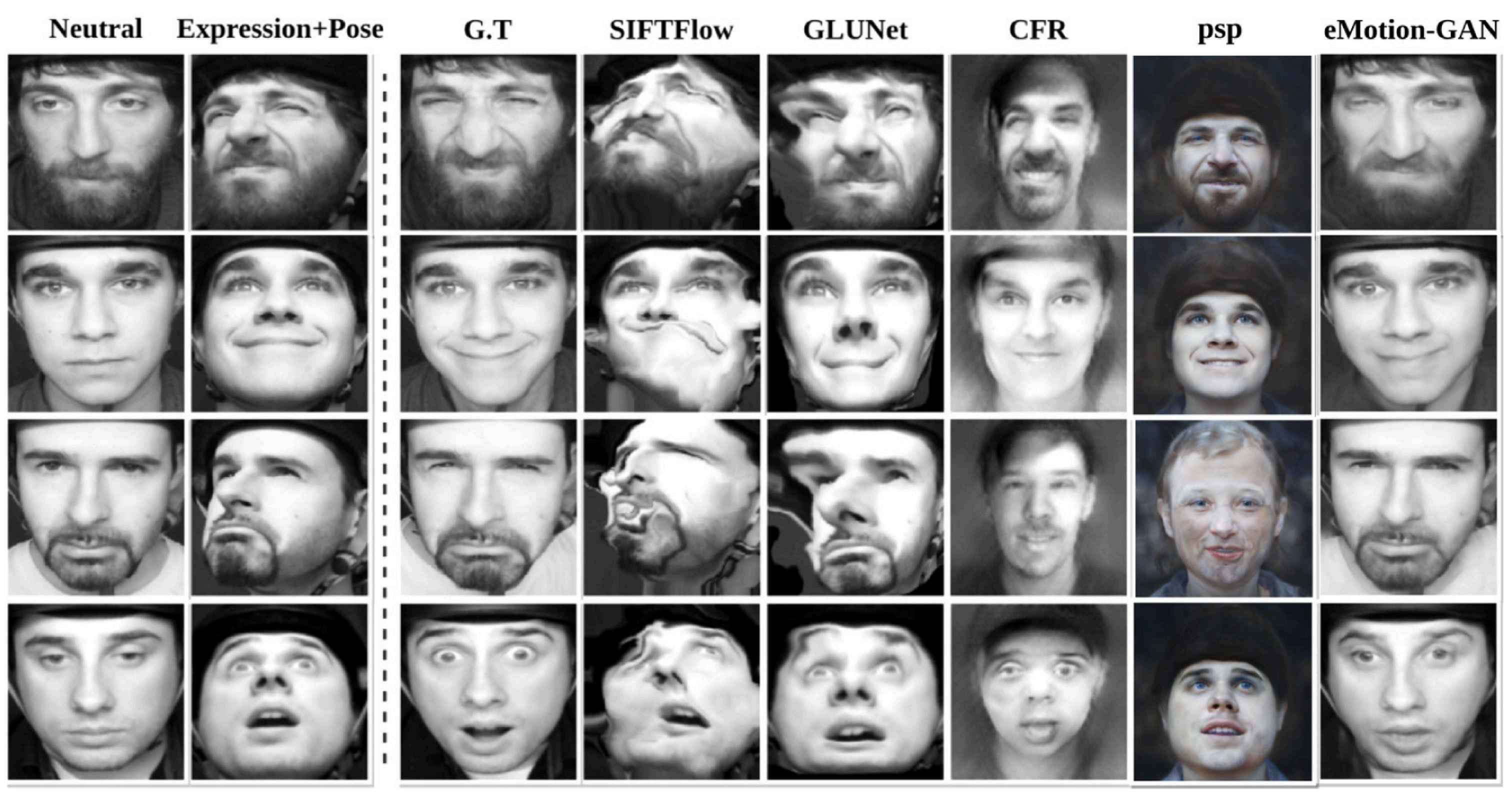
GAN Design

- Utilizes a motion-based discriminator to enforce realism
- Trains to minimize distortion while maximizing expression fidelity
- No Landmarks Needed: Unlike many frontalization methods, performance doesn't depend on facial landmark detection accuracy.



Evaluation & Results

- **Frontal View Synthesis & Expression Preservation**



Method	EPE ↓	SSIM ↑	RMSE ↓	IT(ms) ↓
SIFT (Liu et al., 2008)	36.07±13.50	0.58±0.13	9.10±0.70	–
CFR (Ju et al., 2022)	44.96±6.24	0.31±0.07	9.99±0.11	13.11
GLUNet (Truong et al., 2020)	33.93±6.58	0.66±0.13	**7.62** ± 1.04	**8.17**
pSp (Richardson et al., 2021)	40.32±6.01	0.36±0.07	9.67±0.10	15.73
Ours	**14.34** ± 3.99	**0.78** ± 0.18	7.75±1.00	11.55

Qualitative evaluation (Mean and standard deviation) .

Performance Improvements

- FER Accuracy: Improved up to +20% over baseline frontalization approaches
- Pose Variation Robustness: Stable results across small to large head pose changes

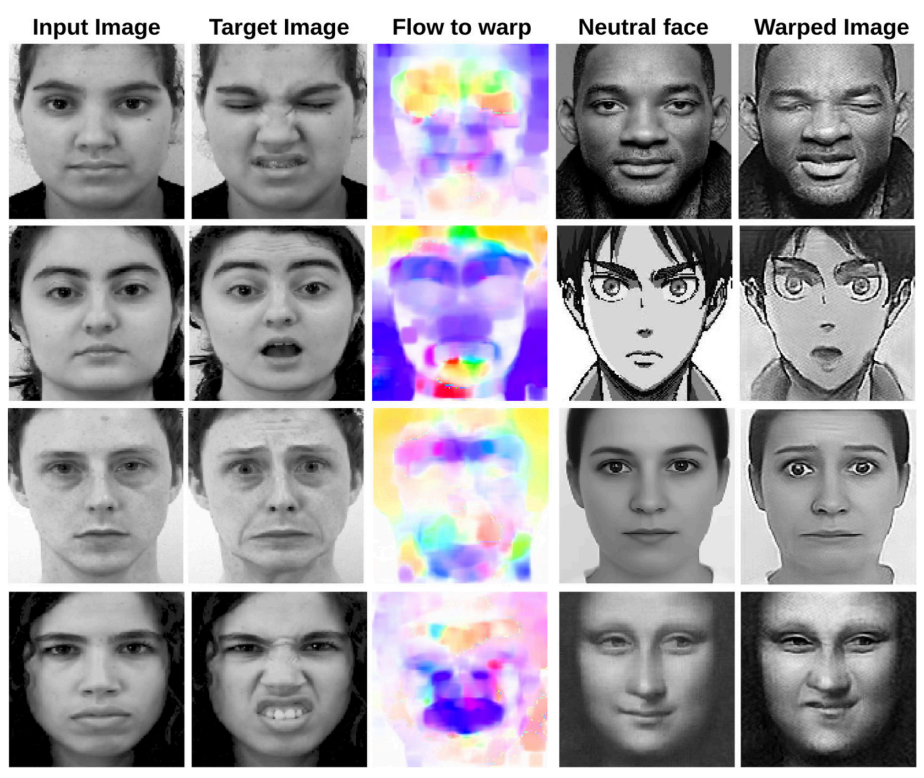
Qualitative Findings

- Synthesized frontal faces show fewer visual artifacts
- Better preservation of subtle expression dynamics such as smiles, frowns, etc

SCAN ME

- Read the Paper
- Explainer Video
- Interactive Demo
- More Results

- **Cross-subject Facial Motion Transfer**



Method	CK+	ADFES	MMI	CASIA	SNAP-F
Ad-Corre	75.33%	85.49%	59.90%	52.92%	55.33%
DAN	71.17%	**93.90%**	60.31%	56.68%	59.85%
HSE	68.33%	89.40%	57.50%	54.61%	54.09%
DMUE	**77.31%**	88.52%	**63.74%**	60.03%	**61.12%**

FER accuracy on warped images using average face.

- Successful transfer demonstrates that expression representation is identity-independent, a key requirement for robust facial analysis systems.
- It validates that the model captures true motion semantics, not just appearance cues.

Limitations & Perspectives

Achievements

- Accuracy improvements ranging from 5% for small pose variations to 20% for extreme poses.

Current Limitations

- May require high-quality optical flow estimation
- GAN training complexity can lead to instability

Future Directions

- Integrate with real-time FER systems
- Extend to multi-view and video frontalization
- Explore self-supervised motion representation learning

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